Tactical Topology: Modelling Football Teams as Communication Networks

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Independent Research · 2026 · github.com/nraptisss/Football-as-a-Communication-Network

Abstract

We recast a football team's passing as a **communication network** and show that metrics from telecommunications and graph theory — fault tolerance, maximum flow and minimum cut, processing latency, and channel interference — map naturally onto recognisable tactical concepts. Using the freely available StatsBomb event data for the entire 2015/16 Spanish La Liga season (380 matches, 760 match–team observations), we build weighted directed passing networks and define four interpretable metrics on them: a node-failure *resilience* score that identifies a team's structurally critical players; a zone-level *max-flow / min-cut* measure of how directly a team progresses the ball; a *tempo* (latency) measure of how quickly players relay the ball; and a *pressing-as-interference* measure of how opponent pressure degrades passing quality. We then layer three deliberately interpretable machine-learning models on the resulting features: unsupervised tactical clustering, supervised match-outcome prediction from first-half structure only, and change-point detection of in-match tactical shifts. The metrics behave as football intuition predicts — the most critical Barcelona players are its midfield and ball-playing defenders, pass completion falls from 81% to 73% under pressure, and direct/counter-attacking sides score higher on progression directness than possession sides. Clustering cleanly separates the elite possession teams from the rest, and first-half network structure predicts the full-time result with 43% cross-validated accuracy against a 37% majority baseline. All code, data and figures are open source and reproducible.

Keywords: football analytics; network science; passing networks; communication networks; max-flow; graph resilience; machine learning; StatsBomb.

1 Introduction

Association football is, at its core, a problem of moving a single token — the ball — through a contested space using a network of eleven cooperating agents. Viewed this way, a team is strikingly similar to a **communication network**: players are nodes that receive, process and forward a signal, and passes are the messages that travel between them. This analogy is more than rhetorical. Decades of work in network science and telecommunications have produced precise, well-understood tools for exactly the questions coaches care about: which nodes are critical and what happens if they fail; how much traffic a network can route from one region to another and where the bottleneck lies; how quickly nodes relay messages; and how external interference degrades throughput.

Passing networks have been studied since the foundational work of Gould and Gatrell [1] and popularised for the modern game by López Peña and Touchette [2] and Buldú and colleagues [3, 4]. Most of this literature characterises network *structure* — centrality, clustering, motifs — and relates it to style or quality. Our contribution is to take the communication-network analogy literally and **operationalise four telecommunications concepts as a coherent, reusable toolkit**, then to validate them against football intuition and use them as features for interpretable machine learning. Figure 1 summarises the mapping.

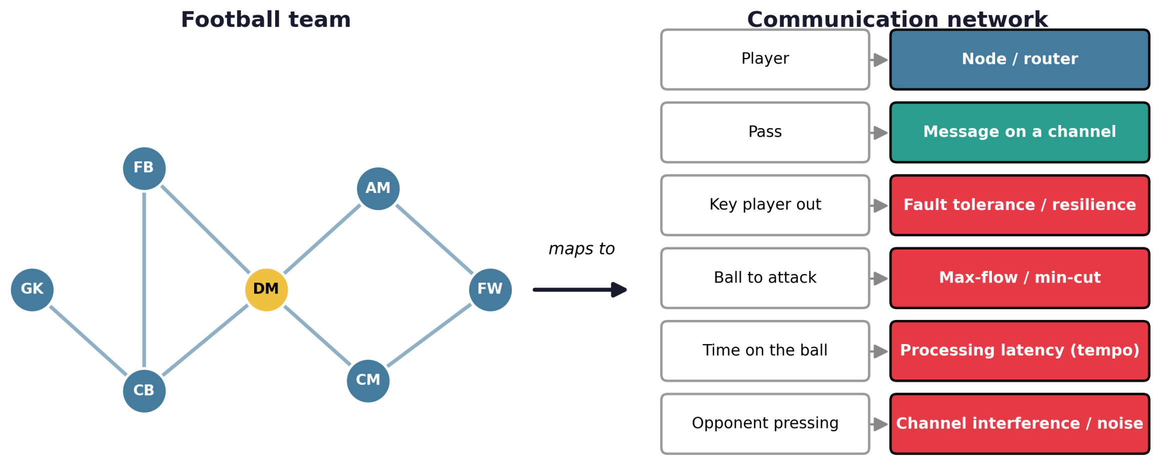


Figure 1. The conceptual mapping at the heart of this work. Each football object on the left has a precise counterpart in communication-network theory on the right; the right-hand concepts (in red) are the four metrics we implement and validate.

Concretely, the paper makes three contributions. (i) We define and implement four communication-network metrics on football passing graphs — resilience, max-flow/min-cut, tempo and pressing-interference — with explicit, parameterised definitions. (ii) We validate each against known football facts on a fixed set of ground-truth matches. (iii) We build three interpretable ML models on top of the metrics and report honest, cross-validated performance on a full league season. The entire pipeline is open source and reproducible from public data with no credentials.

2 Related Work

Passing networks. Gould and Gatrell [1] first analysed a football match as a network. López Peña and Touchette [2] formalised weighted, directed passing networks for the 2010 World Cup and used centrality to rank players and probe the effect of removing them. Buldú et al. [3, 4] conducted an extensive network-science study of Guardiola's Barcelona, reporting its unusually high clustering and balanced flow across pitch zones. Gyarmati et al. [5] characterised team style through *flow motifs* (recurring three- and four-pass sub-sequences), and Cintia et al. [6] linked network indicators to team performance.

Robustness. Closest to our resilience metric, Ichinose et al. [7] removed nodes and links from J-League passing networks and found that, like scale-free communication networks [12], they are robust to random failure but vulnerable to targeted attack — the network-science framing we adopt directly. **Flow and tempo.** Bekkers and Dabadghao [8] studied passing flow at the motif level; we instead aggregate flow spatially and apply the classical max-flow/min-cut theorem [13] to locate progression bottlenecks. **Prediction.** A growing body of work feeds network features to machine-learning models to predict match outcomes [9]; we follow this line but deliberately restrict ourselves to interpretable models and first-half-only features.

Data. Open event datasets such as Pappalardo et al. [10] and StatsBomb Open Data [11] have made this research reproducible; we use the latter. Relative to prior work, our novelty is the *unification* of fault-tolerance, max-flow, latency and interference as a single communication-network toolkit, validated metric-by-metric and wired into an interpretable ML layer.

3 Data

We use StatsBomb Open Data [11] for the 2015/16 Spanish La Liga season (competition 11, season 27): 380 matches with full event data, two teams each, giving 760 match–team observations. Events are accessed through the `statsbombpy` library, which requires no credentials. From the raw events we retain passes and apply a fixed cleaning contract: drop passes with missing origin or destination coordinates; keep an explicit integer team identifier alongside team name; restrict to the first and second half by default; convert the per-period `HH:MM:SS` timestamp to a continuous match clock in seconds; keep incomplete passes (they matter for resilience) but flag completion as a Boolean; and coerce the under-pressure flag to Boolean. For deterministic validation we fix five ground-truth matches spanning Barcelona, Atlético Madrid and Real Madrid, used throughout for sanity checks and unit tests.

4 Methods

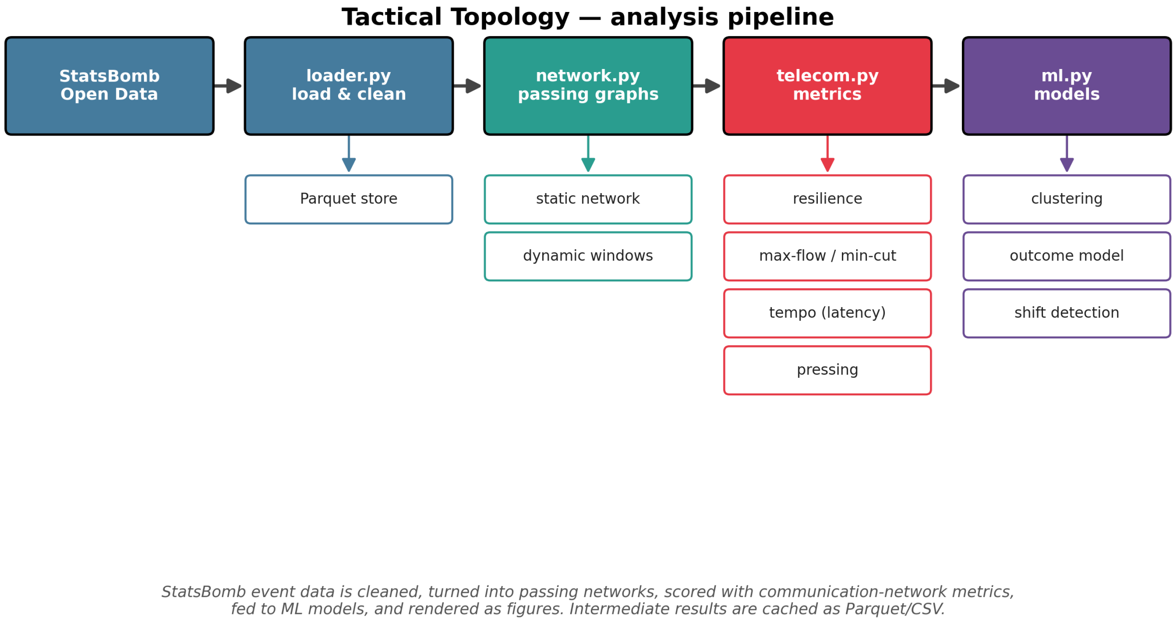


Figure 2. The analysis pipeline. Cleaned events become passing networks, which are scored by the four communication-network metrics and fed to three machine-learning models; all intermediate artefacts are cached.

4.1 Passing-network construction

For one team in one match we build a weighted directed graph in which nodes are players and a directed edge from passer to receiver carries a weight equal to the number of *completed* passes between that ordered pair. Self-passes are removed and substitutes are retained as nodes so that their limited involvement is reflected rather than hidden. We use the StatsBomb pitch coordinate system (120×80) unchanged, placing each node at the player's mean pass-origin location. A *dynamic* variant slides a five-minute window in one-minute steps to produce a time series of graphs. For each graph we summarise density, average clustering, and three complementary 'key player' views: betweenness [14] (the bridging router), weighted eigenvector centrality [15] (a hub among hubs), and weighted PageRank [16] (a player fed by influential team-mates). Graphs are built with NetworkX [17].

4.2 Communication-network metrics

Resilience (fault tolerance). Inspired by attack-tolerance analysis of complex networks [7, 12], we remove each player in turn and measure the resulting degradation in network density, average clustering, and the number of weakly connected components (fragmentation). Each impact is min-max normalised across players and averaged, together with the player's centrality, into a single score in [0, 1]:

$$S(v) = mean[\ norm(\Delta density),\ norm(\Delta clustering),\ norm(\Delta components),\ norm(centrality)\ ]$$

A higher score marks a more structurally critical player. **Max-flow / min-cut (throughput).** We coarse-grain the pitch into a 5×3 grid and build a zone graph whose edge weights count completed passes between zones. Adding a super-source over the three defensive-third zones and a super-sink over the three attacking-third zones, the max-flow/min-cut theorem [13] gives the maximum ball throughput from defence to attack and the bottleneck zones that limit it. We define *attacking flow efficiency* as the max-flow value divided by total completed passes — a measure of progression *directness per pass*.

Tempo (latency). Treating each player as a processing node, we measure the time between receiving a pass and playing the next one. For every reception we take the player's next completed pass *in the same period* and keep the delta if it is at most 30 seconds, guarding against the half-time discontinuity in the continuous clock. We report per-player mean and median touch time, a low-latency ratio (fraction of touches under two seconds), and a touch-weighted team tempo. **Pressing as interference.** We split a team's passes into under-pressure and pressure-free sub-networks and compare their summaries, reporting the density degradation and the drop in completion rate — the signal loss induced by opponent 'noise'.

4.3 Machine-learning models

For every match and team we assemble a feature vector of twelve numeric features: density, average clustering, the value of the top betweenness, eigenvector and PageRank nodes, node and edge counts, attacking flow efficiency, team tempo, low-latency ratio, and the two pressure-degradation measures. Crucially we store centrality *values*, not the player-name strings, so that the features are model-ready; names are kept only for display. Missing values (e.g. when a team has no under-pressure passes) are median-imputed. We then train three models with scikit-learn [18]. *Tactical clustering* applies k-means (k = 4) to the season-average feature vector of each team after standardisation — standardisation is essential because the features live on very different scales. *Outcome prediction* trains a random forest to predict the full-time result (win/draw/loss) from **first-half features only**, evaluated with stratified cross-validation; using only the first half makes the task harder and more interesting than using the whole match. *Tactical-shift detection* z-scales the [density, top-betweenness, node-count] vector of each dynamic window and flags timestamps where the cosine distance between consecutive windows exceeds the mean plus a threshold times the standard deviation.

5 Results

5.1 Network structure

The constructed networks have the expected shape: 13–14 nodes per team, densities from about 0.28 for low-possession sides up to 0.64 for Barcelona, and a clear high-betweenness router. Figure 3 shows Barcelona's network in the 4–0 win at Real Madrid, with Iniesta — the top-betweenness node that match — highlighted. The layout reproduces a recognisable possession shape rather than a featureless blob.

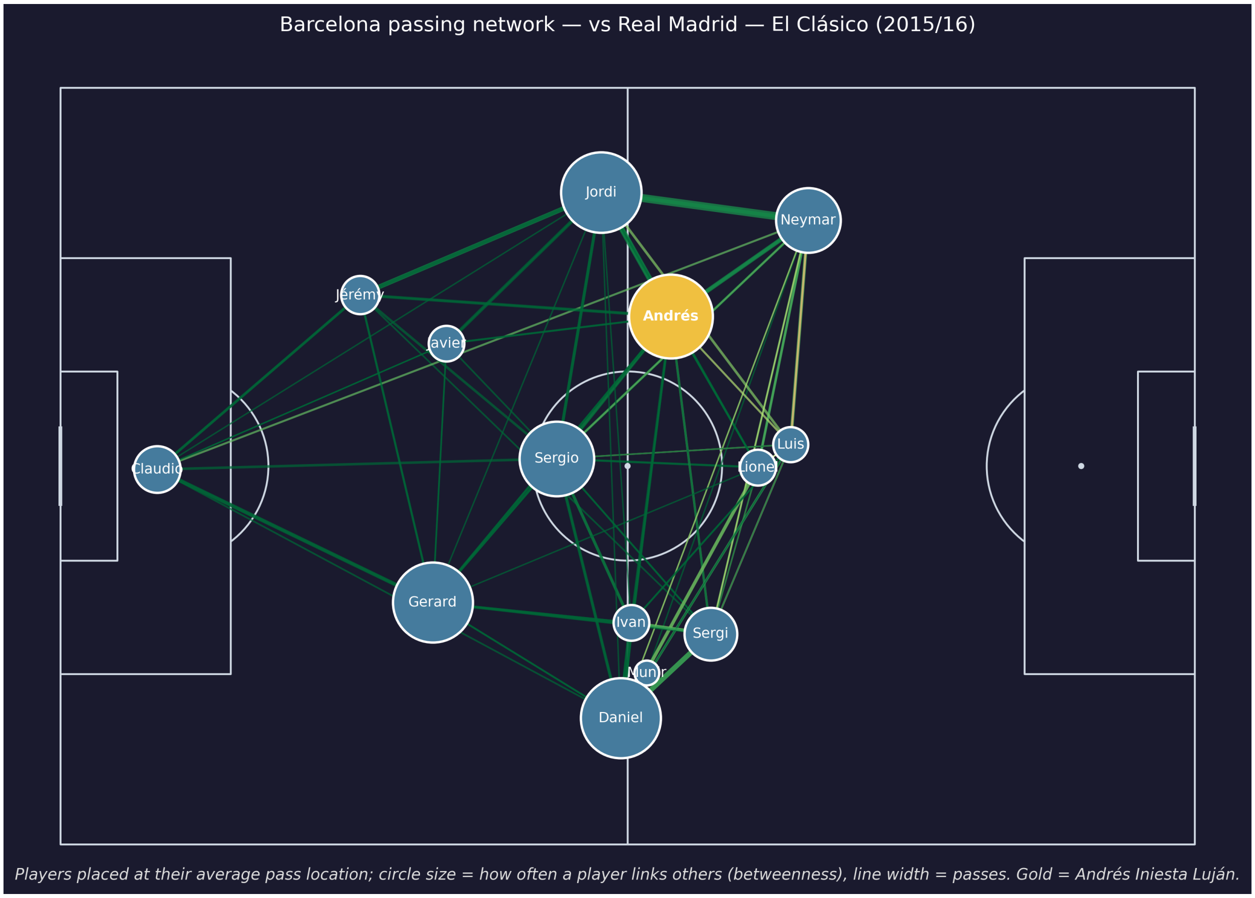


Figure 3. Barcelona's passing network (El Clásico, 2015/16) drawn on the pitch. Node size is betweenness centrality, edge width is pass volume, edge colour is completion rate; gold marks the highlighted router (Iniesta).

5.2 Resilience

The resilience metric passes its sanity check decisively. For Barcelona, the three most critical players are Alves, Iniesta and Busquets — a ball-playing full-back and the two midfield organisers — rather than orthodox centre-backs, exactly the profile expected of a possession team (Figure 4). This mirrors the targeted-attack vulnerability reported for real passing networks [7].

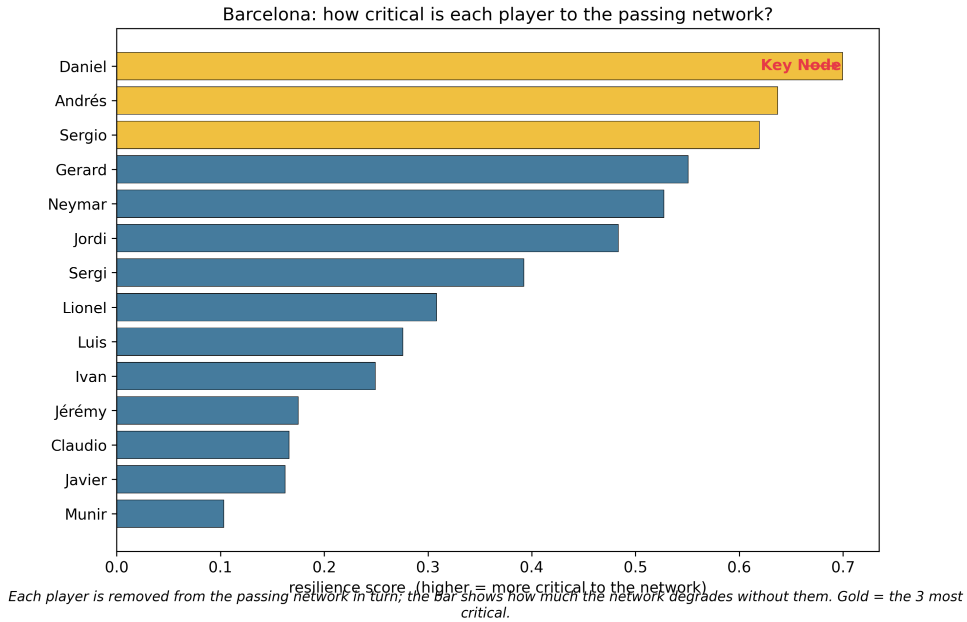


Figure 4. Resilience score per Barcelona player (El Clásico). The three most critical nodes are gold; removing them degrades the network most.

5.3 Flow and directness

Attacking flow efficiency behaves as a measure of *directness*, not of attacking quality. Direct and counter-attacking sides score highest, while possession Barcelona (0.0407) sits below direct teams such as Atlético (0.0446) and well below the most direct sides in the league. This is the correct, expected behaviour: possession teams recycle many lateral passes, inflating the denominator, so their progression per pass is lower. Figure 5 contrasts the two styles on the same matchday; the red band marks the defensive-third min-cut the ball must funnel through.

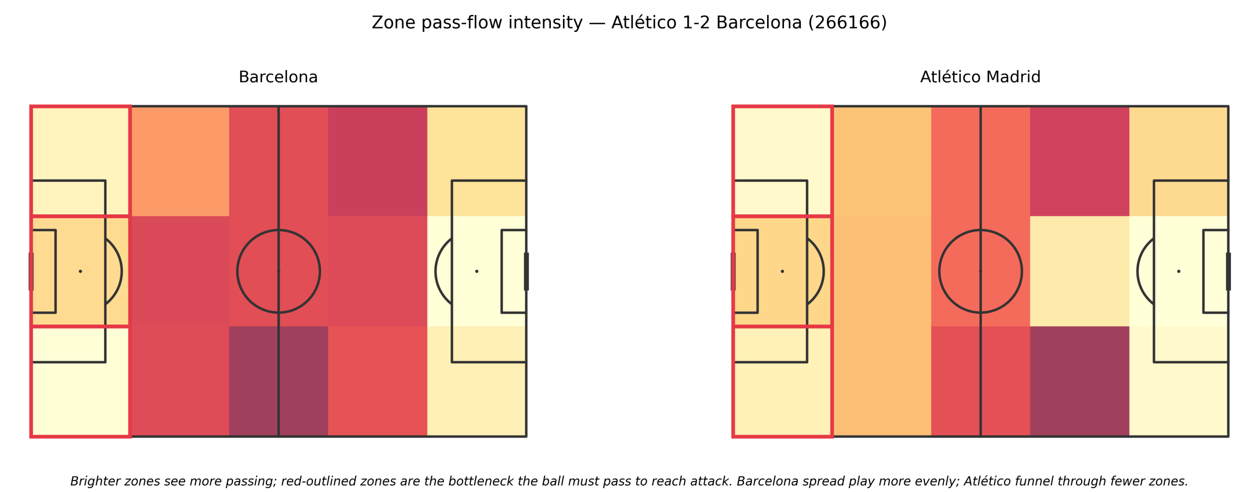


Figure 5. Zone pass-flow intensity for Barcelona and Atlético in the same match. Brighter zones see more passing; the red-outlined zones are the defence-to-attack bottleneck (min-cut).

5.4 Pressing degradation

Pressure clearly degrades the passing signal. Pooled across the ground-truth matches, completion falls from 81% when free to 73% under pressure — a drop of 8.3 percentage points — reproducing a well-known football fact and confirming the pressure flag behaves correctly. The radar chart in Figure 6 shows the under-pressure sub-network (red) shrinking inside the free network (blue) on density, clustering and connectivity.

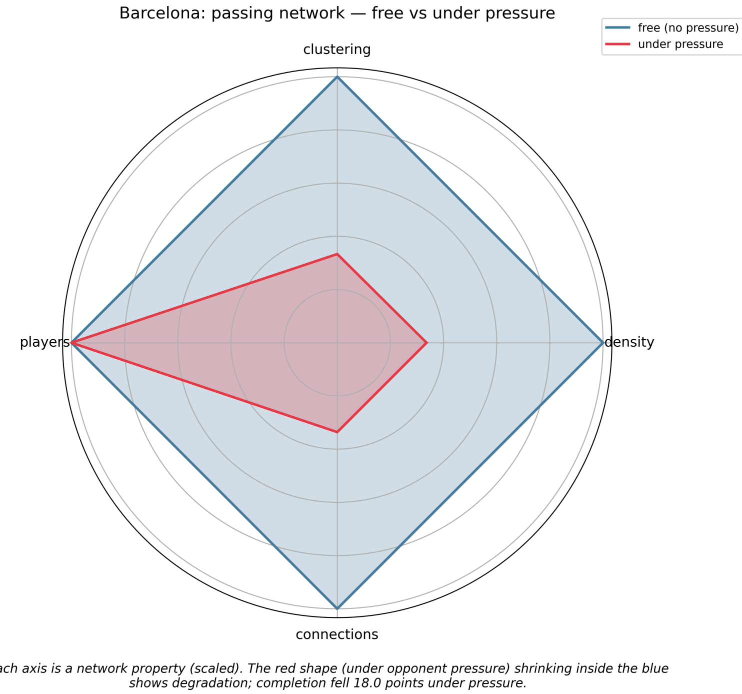
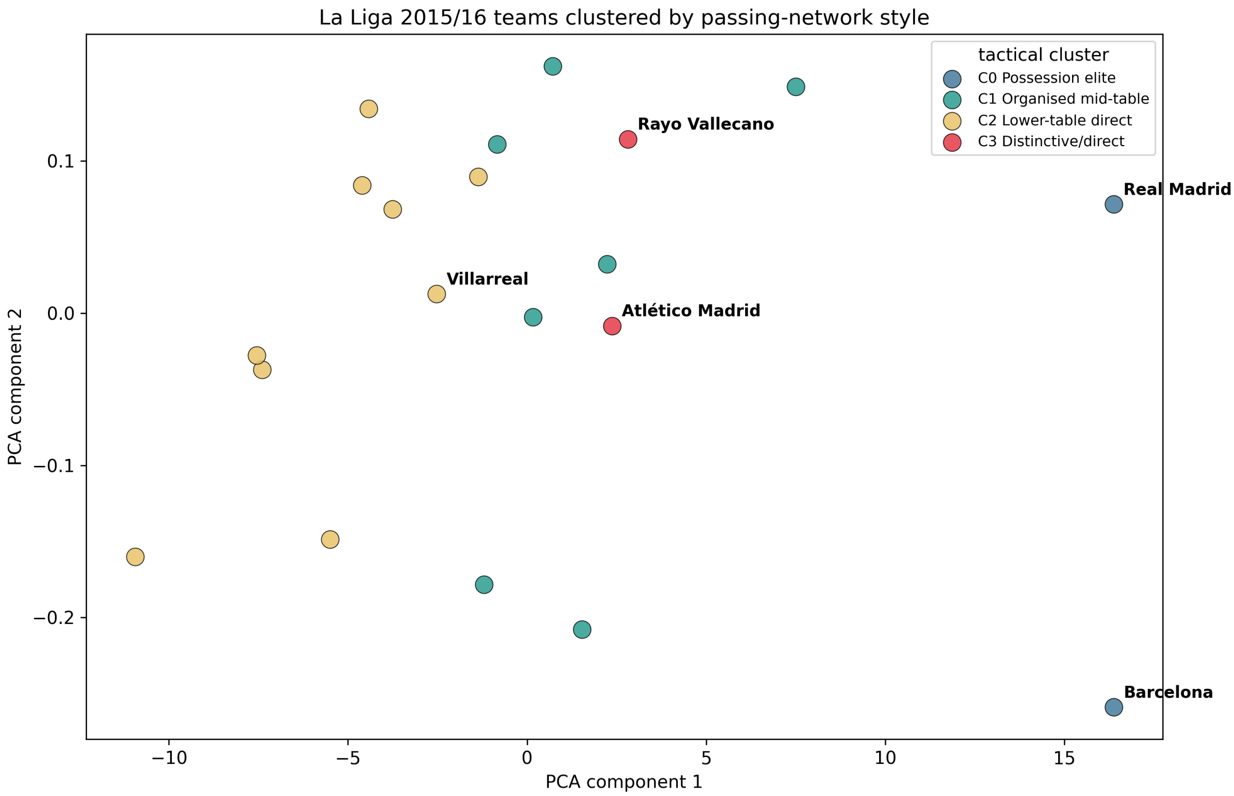


Figure 6. Barcelona's passing network free versus under pressure. The red (pressed) shape sitting inside the blue (free) shape visualises the interference-driven degradation.

5.5 Tactical clustering

Clustering the season-average feature vector of each team into four groups (Figure 7) produces tactically coherent tiers. Cluster 0 isolates the two possession giants (Barcelona, Real Madrid); cluster 3 is a distinctive direct group (Atlético Madrid, Rayo Vallecano); the remaining clusters split the rest of the league into an organised mid-table band and a lower-table direct band. Barcelona and Atlético land in different clusters, as football intuition demands.



Each point is a team; nearby teams play structurally similar football. Colours are the four tactical clusters found automatically from network metrics.

Figure 7. Teams projected to two PCA dimensions and coloured by tactical cluster. Villarreal (annotated) is a mild outlier — its network metrics, not its league finish, place it in the lower-table-direct cluster.

5.6 Match-outcome prediction

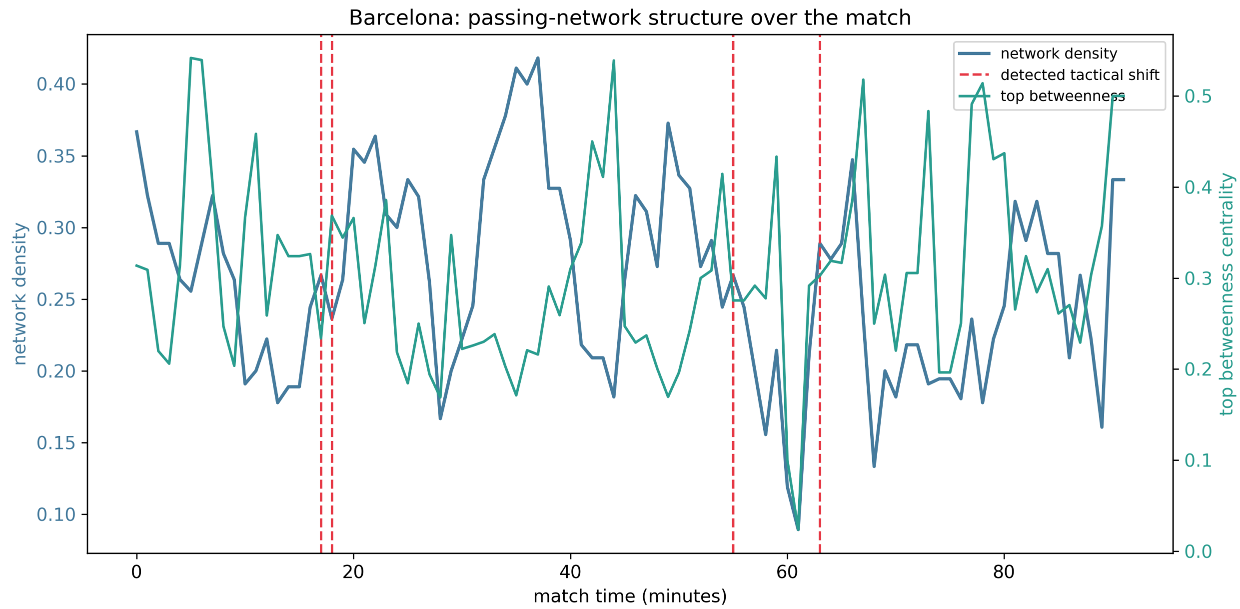
Predicting the full-time result from first-half network structure alone reaches 43% ± 4% cross-validated accuracy with a weighted F1 of 0.413, against a 37% majority-class baseline and a 33% random baseline (Table 1). For a deliberately interpretable model fed only structural first-half features, beating the majority baseline is a meaningful and honest result. Feature importances are spread fairly evenly, led by average clustering, attacking flow efficiency and team tempo — structure, directness and speed — which is reassuringly sensible.

Metric	Value
Cross-validated accuracy	43.8%
Accuracy std. (folds)	4.2%
Weighted F1	0.413
Majority baseline	37.9%
Random baseline (3-class)	33.3%
Top features	avg. clustering, flow efficiency, team tempo

Table 1. Match-outcome model performance (random forest, first-half features, stratified cross-validation).

5.7 Tactical-shift detection

On a manually inspected match (Barcelona vs Sevilla), all 4 of 4 detected change-points fall within three minutes of a substitution or goal (Figure 8). Detection quality varies by match — an unsupervised detector also flags genuine phase changes that are not subs or goals — which is why such validation is done on a human-chosen match.



How connected the team's passing is, minute by minute. Red dashed lines mark automatically detected changes in tactical structure.

Figure 8. Barcelona's network density and top betweenness over the match. Red dashed lines are automatically detected tactical shifts; grey dotted lines are substitutions and goals.

6 Discussion

The central finding is that the communication-network analogy is not merely evocative but *operationally faithful*: each borrowed metric lands on the player or pattern that football experts would nominate. Fault-tolerance analysis surfaces the metronomes and ball-playing defenders; max-flow recovers the intuition that possession and directness are distinct axes; latency quantifies tempo; and interference captures the cost of pressing. Because every feature has a concrete meaning, the downstream models are interpretable by construction — a clustering tier or an outcome prediction can always be traced back to a tactical quantity. The flow-efficiency result is a good example of the analogy doing useful work: it forces a precise definition (throughput per pass) that cleanly separates two styles which a vague notion of 'attacking' would conflate.

7 Limitations

The study is descriptive, not causal: network structure correlates with outcomes but we do not claim it produces them. We analyse a single league season, and several validations (resilience, shift detection) are inspected on a small ground-truth set rather than scored at scale. The defensive-third min-cut is often the same band across teams, so the heatmap's value lies more in the flow intensities than in the cut location. The continuous match clock assumes fixed 45-minute halves, which is harmless for within-period deltas but should not be read across half-time. Finally, the outcome model is honestly modest; richer features (possession value, expected goals) would likely help but were deliberately excluded to keep the study network-pure.

8 Conclusion

Treating a football team as a communication network yields a small set of precise, interpretable metrics — resilience, max-flow, tempo and interference — that align with expert intuition and serve as honest features for tactical clustering, outcome prediction and shift detection. The approach is fully reproducible from open data. Future work includes multi-season and multi-league validation, possession-value and expected-goals integration, and a multilayer treatment that combines passing with off-ball movement.

Acknowledgements

This work is built entirely on StatsBomb Open Data [11]; we thank StatsBomb for making it freely available. Analysis uses NetworkX [17] and scikit-learn [18].

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